

Agentic AI Problem Statement

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VISUALIZE'26

Problem Statement:

Many citizens are unaware of the government schemes they are eligible for due to the large number of central and state welfare programs, complex eligibility criteria, and scattered information sources. There is a need for an intelligent, autonomous AI agent that can understand user requirements, analyze eligibility conditions, and provide personalized guidance for applying to suitable schemes in a simple and accessible manner..

Short Description:

The Autonomous Scheme Navigator Agent is an AI-powered system designed to help citizens discover and apply for relevant government schemes. The agent understands user inputs, matches eligibility criteria with available central and state schemes, and provides step-by-step guidance throughout the application process. It also handles follow-up queries and updates recommendations based on user responses.

Long Description:

The Autonomous Scheme Navigator Agent is an intelligent, goal-driven AI system designed to simplify and streamline citizens' access to government welfare schemes and benefits. Many individuals struggle to identify relevant central and state government programs due to scattered information sources, complex eligibility conditions, technical language, and frequent policy updates. This AI agent acts as a smart digital assistant that understands user requirements through natural language interaction and autonomously processes their details such as age, income, occupation, location, education level, social category, disability status, and other relevant criteria. Using advanced natural language processing and reasoning capabilities, the agent searches through a structured and continuously updated database of government schemes, analyzes eligibility rules, and matches users with the most suitable programs. Beyond simple recommendation, the system provides personalized explanations of benefits, required documents, deadlines, and direct application procedures in clear and simple language. It can handle follow-up questions, refine recommendations based on additional user inputs, and maintain conversational context for a seamless experience. The agent may integrate with official government portals through APIs or automated data retrieval mechanisms to ensure up-to-date and accurate information. By functioning autonomously to plan, search, match, and guide users step-by-step, the system enhances accessibility, transparency, and efficiency in public service delivery, ultimately reducing the gap between citizens and government welfare opportunities.

Expected Outcomes:

- A functional AI agent capable of understanding user requirements in natural language.
- Accurate matching of users with relevant central and state government schemes.
- Personalized recommendations based on eligibility criteria.
- Step-by-step guidance for application procedures.
- Ability to handle follow-up queries and dynamic updates.
- Improved awareness and accessibility of government benefits among citizens

**Problem Statement:**

Hospitals face significant challenges in managing complex workflows such as patient scheduling, resource allocation, staff coordination, and emergency response management. Inefficient scheduling, uneven resource utilization, delayed decision-making, and lack of real-time coordination can lead to long waiting times, staff burnout, increased operational costs, and compromised patient care. There is a need for an intelligent, adaptive AI agent that can optimize hospital operations by analyzing historical data, monitoring real-time conditions, and autonomously supporting decision-making to improve efficiency and patient outcomes..

Short Description:

The Hospital Workflow Agent is an intelligent AI-driven system designed to streamline hospital operations by optimizing patient scheduling, managing resources, coordinating staff, and supporting emergency response activities. The agent learns from historical hospital data and adapts to real-time situations to enhance operational efficiency and improve the quality of patient care..

Long Description:

The Hospital Workflow Agent is an intelligent AI-based system designed to optimize hospital operations by improving patient scheduling, resource allocation, staff management, and emergency response coordination. Hospitals often face challenges such as long waiting times, inefficient use of beds and medical equipment, staff workload imbalance, and delays during emergencies. This agent analyzes historical data including patient admissions, treatment durations, peak hours, staff schedules, and equipment usage to identify patterns and predict future demand. Using machine learning and real-time monitoring, it dynamically adjusts appointment schedules, allocates beds and operation theaters efficiently, balances staff shifts, and prioritizes critical cases during emergencies. The system integrates with hospital management systems and electronic health records while ensuring data security and regulatory compliance. By continuously learning and adapting to real-time situations, the Hospital Workflow Agent enhances operational efficiency, reduces costs and delays, supports healthcare professionals, and ultimately improves patient care and overall hospital performance.

Expected Outcomes:

- An AI agent capable of optimizing patient scheduling and reducing waiting times.
- Improved allocation and utilization of hospital resources such as beds and equipment.
- Efficient staff scheduling and workload balancing.
- Real-time emergency response coordination and prioritization.
- Data-driven decision support for hospital administrators.
- Enhanced operational efficiency and reduced costs.
- Improved patient satisfaction and overall healthcare outcomes.

**Problem Statement:**

Traditional learning systems often fail to maintain student engagement, motivation, and personalized support. Many learners struggle due to one-size-fits-all teaching approaches, lack of adaptive learning paths, and minimal feedback mechanisms. Additionally, the absence of interactive and motivating elements can reduce interest and consistency in learning. There is a need for an intelligent AI-based mentor agent that personalizes learning experiences, adapts to individual learning styles, tracks progress, identifies knowledge gaps, and uses gamification techniques to enhance motivation and academic performance.

Short Description:

The Gamified Learning Mentor Agent is an AI-powered personalized tutoring system that adapts to individual learning styles and creates customized learning paths. By incorporating gamification elements such as points, badges, levels, and challenges, the agent keeps learners engaged while tracking progress, identifying weaknesses, and providing targeted support to improve learning outcomes.

Long Description:

The Gamified Learning Mentor Agent is an intelligent AI-powered tutoring system designed to deliver personalized and engaging learning experiences through adaptive learning and gamification techniques. Traditional learning methods often fail to address individual differences in learning pace, style, and understanding, leading to reduced motivation and performance. This AI mentor analyzes each learner's knowledge level, strengths, weaknesses, and preferences to create customized learning paths tailored to their needs. It continuously tracks progress, monitors performance through quizzes and assessments, and identifies knowledge gaps to provide targeted interventions and instant feedback. To enhance motivation and engagement, the agent incorporates game mechanics such as points, badges, levels, leaderboards, rewards, and interactive challenges, encouraging consistent participation and goal-oriented learning. The system dynamically adapts in real time based on learner responses and performance trends, ensuring effective knowledge retention and skill development. By combining personalization, continuous assessment, and gamified elements, the Gamified Learning Mentor Agent improves engagement, boosts academic performance, and makes learning more interactive and enjoyable.

Expected Outcomes:

- A personalized AI tutor that adapts to individual learning styles and pace.
- Customized learning paths based on learner performance and preferences.
- Integration of gamification elements such as points, badges, levels, and challenges.
- Continuous progress tracking and knowledge gap identification.
- Targeted interventions and instant feedback for improved understanding.
- Increased learner engagement, motivation, and academic performance.



Problem Statement:

Startups often face financial instability due to limited resources, unpredictable cash flow, poor expense tracking, and lack of financial expertise. Founders frequently struggle to monitor budgets, forecast revenue, manage operational costs, and prepare investor reports effectively. Manual financial management methods are time-consuming and prone to errors, which can lead to poor decision-making and financial risk. There is a need for an intelligent autonomous AI agent that can manage startup finances efficiently, provide real-time financial insights, and support data-driven decision-making to ensure sustainable growth..

Short Description:

The Startup Finance Management Agent is an AI-powered financial assistant designed to help startups manage expenses, forecast cash flow, optimize budgets, and generate investor-ready reports. The agent analyzes financial data, monitors key performance indicators, and provides actionable insights and alerts to maintain strong financial health.

Long Description:

The Startup Finance Management Agent is an intelligent AI-powered system designed to help startups efficiently manage their finances and make data-driven decisions. Startups often face challenges such as irregular cash flow, uncontrolled expenses, limited financial expertise, and difficulty in preparing investor reports. This autonomous agent integrates with accounting tools and financial data sources to automatically track expenses, categorize transactions, monitor revenue, and maintain accurate financial records. Using machine learning and predictive analytics, it forecasts cash flow, estimates future income and expenditures, and identifies potential liquidity risks in advance. The agent optimizes budgets by analyzing spending patterns and suggesting cost-saving strategies, while also generating real-time alerts for overspending, unusual transactions, or low cash reserves. Additionally, it creates automated financial dashboards and investor-ready reports such as profit and loss statements and key performance metrics. By continuously analyzing financial health indicators and adapting to changing business conditions, the Startup Finance Management Agent reduces manual workload, improves financial transparency, supports strategic planning, and ensures sustainable growth for startups.

Expected Outcomes:

- Automated expense tracking and financial record management.
- Accurate cash flow forecasting and financial trend prediction.
- Optimized budget planning with cost-saving recommendations.
- Real-time alerts for financial risks and health indicators.
- Generation of investor-ready financial reports and dashboards.
- Improved financial transparency and data-driven decision-making.
- Enhanced financial stability and sustainable startup growth.

**Problem Statement:**

Organizations handle numerous contracts across vendors, clients, employees, and partners, each containing complex terms, obligations, deadlines, and compliance requirements. Manual contract tracking is time-consuming, error-prone, and often leads to missed deadlines, financial penalties, compliance risks, or legal disputes. There is a need for an intelligent AI-based agent that can automatically analyze contracts, extract critical information, monitor obligations, assess risks, and provide timely alerts to ensure compliance and reduce operational and legal risks.

Short Description:

The Contract Monitoring Intelligence Agent is an AI-powered system designed to automatically parse contracts, extract key clauses and deadlines, monitor compliance, and alert stakeholders about upcoming obligations or potential breaches. It supports multiple contract types and provides risk assessment insights to improve contract management efficiency.

Long Description:

The Contract Monitoring Intelligence Agent is an AI-powered system designed to automate and streamline contract management by analyzing and monitoring contractual obligations across multiple contract types such as vendor, service, employment, lease, and partnership agreements. Using Natural Language Processing (NLP) and machine learning, the agent parses contracts to extract key terms including duration, renewal dates, payment schedules, performance obligations, penalty clauses, compliance requirements, and termination conditions. It continuously tracks these obligations, monitors compliance, and generates real-time alerts for upcoming deadlines, renewals, payments, or potential breaches. The system also performs risk assessments by identifying ambiguous clauses, unfavorable terms, financial liabilities, and non-compliance risks, providing stakeholders with actionable insights and summary dashboards. Integrated with enterprise systems and secure document storage, the agent ensures data privacy and regulatory compliance. By automating extraction, monitoring, alerting, and risk analysis, it reduces manual workload, prevents missed commitments, minimizes legal and financial risks, and improves overall contract lifecycle management efficiency.

Expected Outcomes:

- Automated parsing and extraction of key contract terms and clauses.
- Support for multiple types of contracts across departments.
- Real-time monitoring of deadlines, obligations, and compliance requirements.
- Automated alerts for renewals, payments, and potential breaches.
- Risk assessment reports highlighting legal and financial exposure.
- Improved efficiency, reduced manual workload, and minimized compliance risks.
- Enhanced contract transparency and better stakeholder decision-making.

**Problem Statement:**

Organizations, startups, and research institutions operate in highly competitive and rapidly evolving environments where staying updated on research trends, patent developments, competitor strategies, and industry news is critical. However, manually monitoring scientific publications, patent databases, news platforms, and social media sources is time-consuming, inefficient, and prone to missing important updates. The lack of timely insights can result in lost opportunities, delayed innovation, and weakened competitive positioning. Therefore, there is a need for an autonomous AI agent capable of continuously tracking research and competitor activities, analyzing vast information sources, and delivering concise, actionable insights in real time.

Short Description:

The Research & Competitor Tracking Agent is an autonomous AI-powered system that continuously monitors scientific publications, patent filings, news articles, and social media platforms to identify research trends and competitor activities. It provides summarized insights, detects emerging opportunities, and alerts users about relevant developments to support strategic decision-making.

Long Description:

The Research & Competitor Tracking Agent is an autonomous AI-powered system designed to continuously monitor scientific publications, patent filings, industry news, blogs, and social media platforms to track research trends and competitor activities. Using Natural Language Processing (NLP), machine learning, and data mining techniques, the agent collects and analyzes large volumes of information to identify emerging technologies, innovation patterns, market shifts, and competitor strategies such as product launches, partnerships, funding announcements, and patent submissions. It extracts relevant data, categorizes insights by industry and relevance, performs trend forecasting and sentiment analysis, and provides concise summaries, dashboards, and periodic analytical reports. While continuously learning from user preferences to improve accuracy and relevance. By automating competitive intelligence and research monitoring, the agent supports proactive decision-making, strategic planning, innovation management, and helps organizations maintain a strong competitive advantage.

Expected Outcomes:

- Continuous automated monitoring of research publications, patents, news, and social media.
- Identification of emerging research trends and technological advancements.
- Tracking of competitor activities including product launches, patents, and partnerships.
- Summarized insights and periodic analytical reports.
- Real-time alerts for critical developments and opportunities.
- Data-driven strategic decision-making support.
- Improved innovation planning and competitive advantage.



Problem Statement:

With rapid advancements in artificial intelligence, there is a growing need for innovative solutions that go beyond simple automation and demonstrate true autonomous decision-making and multi-step reasoning. However, many existing AI applications are limited to single-task responses and lack the ability to independently plan, interact with external tools, and execute meaningful real-world tasks. This open innovation track aims to encourage students to design and develop novel Agentic AI applications that showcase autonomy, intelligent reasoning, and practical impact across diverse domains.

Short Description:

The Student Innovation – Open Agentic AI Track invites participants to propose and build their own innovative AI agent capable of autonomous decision-making, multi-step reasoning, and interaction with external tools or APIs. The solution should address a real-world problem and demonstrate the practical capabilities of Agentic AI systems.

Long Description:

The Student Innovation – Open Agentic AI Track is an open innovation category that encourages participants to design and develop a novel Agentic AI solution capable of solving real-world problems through autonomous decision-making and multi-step reasoning. Unlike traditional AI systems that respond to single prompts, the proposed solution must demonstrate goal-oriented behavior, the ability to break down complex tasks into sequential steps, adapt to dynamic inputs, and interact with external tools, APIs, databases, cloud services, or other systems to accomplish meaningful objectives. Participants can choose any domain such as healthcare, education, finance, sustainability, governance, business, or technology, provided the solution shows clear problem identification, intelligent planning, contextual memory handling, real-time adaptation, and actionable outputs. The agent should function as a scalable and practical prototype that highlights innovation, usability, measurable impact, and responsible AI practices including data privacy and security..

Expected Outcomes:

- Development of a novel and innovative Agentic AI solution.
- Demonstration of autonomous decision-making and multi-step reasoning.
- Integration with external tools, APIs, or systems to accomplish tasks.
- Clear problem identification and practical real-world impact.
- Scalable and implementable prototype or working model.
- Consideration of ethical, privacy, and security aspects.
- Enhanced understanding of Agentic AI design and deployment.